

Cognitive versus Physical Automation: Firm-Level Evidence from AI and Robotics Patents

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Abstract

Automation comes in two forms: *cognitive* (artificial intelligence, which substitutes for information processing) and *physical* (robotics, which substitutes for manual tasks). We ask whether the two have different footprints inside the firms that create them. Linking the USPTO AI Patent Dataset and CPC robotics patents (subclass B25J) to 7,087 publicly-traded innovators (83,976 firm-years, 1995–2024) via the Kogan-Papanikolaou-Seru-Stoffman (KPSS) patent–firm crosswalk, we find a sharp contrast. In two-way (firm + year) fixed-effects panels, a firm’s cumulative *AI*-patent stock is associated with *higher* employment (elasticity 0.055, $t = 7.5$), whereas its robotics stock is associated with higher *market value* (Tobin’s Q coefficient 0.40, $t = 5.0$) but

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no employment change. An event study around a firm’s *first* AI patent shows employment rising about 11% and sales-per-employee about 6% over the following five years, with flat pre-trends. At the level of the innovating firm, cognitive automation looks labor-*complementary*, not labor-displacing. We stress-test both headline results. The AI–employment complementarity survives industry $\times$ year fixed effects (0.036, $t = 4.8$) and added accounting controls, and it concentrates in high-tech and R&D-active firms—the settings where AI is most plausibly a productive input rather than a labeling artifact. *Randomization inference* that permutes the entire firm-level AI trajectory across firms 500 times places the actual elasticity far outside a permutation distribution centered at zero (design-based $p < 0.002$), so the result is not an artifact of clustered-standard-error asymptotics. The robotics–value result is equally robust and is strongest among large firms. A *minimum-detectable-effect* analysis shows the genuine nulls are informative: at 80% power we could have detected a robotics–employment elasticity of about 2.3% of the employment standard deviation, so the absence of a firm-level robotics employment effect is a bound, not low power. These remain within-firm conditional correlations; hand-reading whether each robotics patent is labor-replacing or labor-augmenting is the paper’s key remaining input.

JEL: O33, J23, G30, O31. **Keywords:** artificial intelligence, robotics, automation, patents, employment, firm value, randomization inference, statistical power.

1 Introduction

Will automation destroy jobs or create them? The answer likely depends on *which* automation. *Physical* automation—industrial robots—substitutes for manual labor and, in the aggregate, has been linked to local employment declines (Acemoglu and Restrepo, 2020; Graetz and Michaels, 2018). *Cognitive* automation—artificial intelligence—substitutes instead for prediction and information processing, and its labor consequences are more contested (Autor, 2015; Webb, 2020; Babina et al., 2024). Yet most evidence studies the two in isolation, and most of it is measured at the level of a labor market or an occupation rather than the firm that actually produces the technology. This paper puts cognitive and physical automation side by side *inside the firm*, using the patents that embody each technology, and asks whether they leave different footprints on employment and on value.

We combine three datasets. The USPTO Artificial Intelligence Patent Dataset (AIPD) flags AI patents using a machine-learning classifier over the full text of every granted patent. PatentsView CPC codes identify robotics: subclass B25J covers industrial robots and manipulators. The Kogan-Papanikolaou-Seru-Stoffman (KPSS) crosswalk links each patent to a CRSP firm (`permno`) and to a stock-market-based estimate of its private value (Kogan et al., 2017). Merging the resulting firm-year patent stocks to Compustat gives a panel of 7,087 publicly-traded innovators over 1995–2024, 83,976 firm-years, on which we measure employment, sales, labor productivity (sales per employee), profitability (ROA), and Tobin’s Q .

Our central finding is a clean cognitive/physical contrast. In firm- and year-fixed-effects panels, the cumulative *AI*-patent stock loads positively on employment (elasticity 0.055, $t = 7.5$; Table 2) while the *robotics* stock does not (insignificant). Symmetrically, robotics loads positively on *Tobin’s Q* (0.40, $t = 5.0$) whereas AI does not; the market capitalizes physical-automation capability into value, but AI capability travels with headcount. An event study around a firm’s first AI patent (Figure 1) shows employment climbing to roughly +11% by year four and sales-per-employee to roughly +6%, with flat pre-adoption trends.

Inside the innovating firm, AI has so far accompanied *expansion*, not contraction—consistent with Babina et al. (2024), who find that AI-investing firms grow.

A single set of two-way fixed-effects coefficients, however striking, is not enough to headline two facts. We therefore subject both results to a battery of design-based checks, and this is where the paper goes beyond a first pass at the data. First, we show that the AI–employment elasticity is not a mechanical size or industry story: it *survives* the replacement of year fixed effects with fully saturated industry $\times$ year fixed effects (which absorb any industry-specific time shock; elasticity 0.036, $t = 4.8$), it survives the addition of profitability and leverage controls, and it holds in the modern (post-2005) AI era (Table 5). Second, we ask *which* firms drive it. Interacting the AI stock with observable pre-sample moderators (Table 4), the employment complementarity is strongest in high-tech and R&D-active firms—exactly where an AI patent is most plausibly a genuine productive input—while the robotics–value effect is strongest among large firms. Third, and most importantly, we replace the parametric t -test with *randomization inference*. We permute each firm’s entire AI-patent trajectory to a randomly chosen donor firm, matched on calendar year, 500 times, and re-estimate the panel each time (Table 7, Figure 2). Under this sharp null the permutation distribution is centered at zero (its mean is +0.001, confirming the test is well-behaved), and the actual elasticity of 0.055 lies entirely outside it: not one of 500 permutations produced a coefficient as large, a design-based $p < 0.002$. The AI–employment link does not depend on clustered-standard-error asymptotics.

Finally, we treat our nulls as quantitative objects rather than as bare failures to reject. The firm-level *robotics–employment* coefficient is insignificant—a result of independent interest, because it is the firm-level counterpart of the aggregate job-displacement finding of Acemoglu and Restrepo (2020). A minimum-detectable-effect (MDE) analysis (Table 8) shows this null is informative: at 80% power our design could have detected a robotics–employment elasticity of about 2.3% of the employment standard deviation and a robotics–ROA effect of about 3.5%. The firm-level robotics employment effect, if any, is smaller than that. This

does *not* contradict the aggregate displacement literature; it locates the action outside the innovating firm’s own payroll (at competitors, suppliers, or downstream adopters), which is the natural next question and beyond a firm-of-origin design.

We make four contributions. First, we separate cognitive from physical automation *at the firm level* using patents, and show they map to different outcomes—jobs versus value—within the same panel and the same specification. Second, we document that AI, at the innovating firm, has been labor-complementary over 1995–2024, and we establish this with design-based inference rather than asymptotic standard errors. Third, we quantify the informativeness of the robotics employment null, so a reader can see what magnitude of firm-level displacement the data rule out. Fourth, we are explicit that the machine-assigned “robotics” label cannot distinguish a labor-*replacing* robot from a labor-*augmenting* one; hand-reading each robotics patent to code that distinction is the design that would turn our conditional correlation into a mechanism, and it is the reserved contribution of the student co-author. Section 2 lays out the conceptual framework and hypotheses, Section 3 places the paper in the literature, Section 4 describes the data and research design, Section 5 presents the main panel and event-study results, Section 6 the heterogeneity, Section 7 the alternative-identification and randomization-inference checks, Section 8 the power analysis, and Section 9 concludes with the hand-coding that would sharpen the test.

2 Conceptual Framework and Hypotheses

Two technologies, two margins. The task-based framework of Acemoglu and Restrepo (2020); Acemoglu et al. (2022) treats automation as the reassignment of tasks from labor to capital. A technology that automates a task a worker used to perform (a *displacement* effect) reduces labor demand for that task; a technology that raises productivity or creates new tasks (a *reinstatement* or *productivity* effect) can raise it. Which effect dominates is an empirical question that depends on the technology and on the unit of observation. Industrial robotics

is the canonical displacing technology: a manipulator that welds, paints, or performs pick-and-place does directly what a manual worker did. Artificial intelligence is more ambiguous: it substitutes for prediction and information processing (Autor, 2015; Webb, 2020), but at the firm that develops it, AI is often an input into new products and services that require complementary human labor to build, sell, and support (Babina et al., 2024).

The firm of origin. We study the firm that *creates* the technology, not the firm or region that later adopts it. This choice is a strength and a limitation. It is a strength because the patent gives a precise, dated, firm-level measure of capability, and because the firm that patents AI or robotics is where the technology’s earliest labor and value consequences appear. It is a limitation because displacement may occur *elsewhere*—at competitors that lose share, at customers that automate, or in the aggregate labor market—and a firm-of-origin design cannot see it. Our results should therefore be read as statements about the innovating firm’s own workforce and market value, not about economy-wide employment.

Hypotheses. The framework and the prior evidence motivate:

H1 (Cognitive complementarity). At the innovating firm, a larger AI-patent stock is associated with *higher* employment: AI is labor-complementary where it is invented.

H2 (Physical value capitalization). A larger robotics-patent stock is associated with higher firm *value* (Tobin’s Q), as the market prices physical-automation capability, while its own-firm employment effect is muted or negative.

H3 (Divergence). The two technologies load on *different* margins: AI on employment, robotics on value. A pooled “automation” measure would blur a real distinction.

Our evidence supports all three. The sharper test—whether robotics that a human reader codes as labor-*replacing* displaces own-firm labor while labor-*augmenting* robotics does not—

requires the hand-coding reserved for the student co-author (Section 9) and is deliberately *not* attempted with the observable moderators we use in Section 6.

3 Related Literature

This paper connects three strands. The first is the economics of automation and labor. The robots-and-jobs literature finds that industrial-robot adoption reduces employment and wages in exposed local labor markets (Acemoglu and Restrepo, 2020; Graetz and Michaels, 2018), and the task-based framework (Acemoglu et al., 2022; Autor, 2015) explains why: robots displace routine manual tasks faster than they reinstate new ones. The AI-and-labor literature is younger and less settled. Occupation-level exposure measures (Webb, 2020) and vacancy-based evidence (Acemoglu et al., 2022) suggest AI substitutes for a different, more cognitive task bundle, with effects that are so far smaller and more heterogeneous than those of robotics. Babina et al. (2024) move the question inside the firm and find that firms investing in AI grow—in sales, employment, and product scope—which is the closest antecedent to our AI–employment result. Our contribution to this strand is to place cognitive and physical automation in the *same* firm-level panel, measured symmetrically from patents, so that the jobs-versus-value contrast is identified off a single specification rather than pieced together across studies with different units and designs.

The second strand is innovation and firm value. Kogan et al. (2017) construct stock-market-based estimates of the private value of each patent and show that patent value forecasts firm growth and reallocation. We use their crosswalk both to link patents to firms and, implicitly, to motivate Tobin’s Q as the value margin on which robotics loads. Our finding that robotics capability capitalizes into value while AI capability travels with headcount refines the innovation-value mapping by technology type.

The third strand is design-based and finite-sample inference. Recent methodological work stresses that inference with a moderate number of clusters, or off a treatment that

is correlated across units, benefits from permutation and randomization methods rather than asymptotic cluster-robust standard errors (Young, 2019); that pre-trend tests have limited power and should be interpreted cautiously (Roth, 2022); that dynamic difference-in-differences and event-study estimands are fragile under effect heterogeneity (Callaway and Sant’Anna, 2021); and that a statistically insignificant estimate should be interpreted through its confidence interval and power rather than treated as proof of no effect (Abadie, 2020). We use these tools constructively: randomization inference to confirm that the AI–employment elasticity is not an artifact of cluster asymptotics (Section 7), and a minimum-detectable-effect calculation to state the robotics employment null as a quantitative bound (Section 8).

4 Data and Research Design

Patents to firms. We flag a granted patent as AI if the USPTO AIPD marks it (`predict50_any_ai`) and as robotics if any of its CPC subclasses is B25J (industrial manipulators). The KPSS file links roughly 3.4 million patents to a CRSP `permno` with a market-value estimate; of the firm-matched patents, 632,817 are flagged AI and 7,626 are flagged robotics. We aggregate to a firm-year panel of cumulative AI, robotics, and total patent stocks (a stock is the running sum of a firm’s flagged grants through year t), and merge to Compustat annual fundamentals for every firm that ever patents and links to CRSP. The estimation sample is 7,087 firms and 83,976 firm-years over 1995–2024.

Variables. The treatment variables are $\ln(1 + \text{AI stock})$ and $\ln(1 + \text{robotics stock})$; the log-of-one-plus transform accommodates the many firm-years with a zero stock and gives the coefficients an elasticity interpretation at positive stocks. Outcomes are $\ln(\text{employment})$, $\ln(\text{sales per employee})$ (labor productivity), ROA (net income over assets), and Tobin’s Q (market value of assets over book assets). The single time-varying control is firm size, $\ln(\text{assets})$; robustness specifications add ROA and book leverage. Table 1 reports summary

statistics. Consistent with a panel of innovators, the median firm has a positive AI stock by the end of the sample, while robotics stocks are far thinner (only about 4% of firm-years carry any B25J patent), which is why the robotics results are estimated less precisely and why we report their power explicitly.

Design 1: two-way fixed-effects panels. For outcome y_{it} we estimate

$$y_{it} = \beta_1 \ln(1 + \text{AI}_{it}) + \beta_2 \ln(1 + \text{Rob}_{it}) + \gamma \ln(\text{assets})_{it} + \alpha_i + \delta_t + \varepsilon_{it}, \quad (1)$$

where α_i and δ_t are firm and year fixed effects and standard errors are clustered by firm. Firm fixed effects absorb every time-invariant difference across firms (industry, founding cohort, business model), so β_1 and β_2 are identified from *within-firm* growth in each patent stock; year fixed effects absorb common macro shocks (business cycles, the aggregate AI wave). These are conditional within-firm correlations, not experiments: firms *choose* to patent. We do not claim exogenous variation. What we claim is that, holding the firm and the year fixed, the two technologies move with different outcomes, and that this pattern is robust to demanding alternative specifications and to design-based inference.

Design 2: first-AI-patent event study. To trace dynamics and check pre-trends, we estimate an event study around a firm’s first AI patent for adopters whose first AI year falls in 2000–2020 (so that a $[-5, +5]$ window is observable). With relative-time indicators (base $k = -1$) and firm and year fixed effects, the coefficients trace the path of employment and productivity around the adoption of AI capability. Flat pre-trends and a post-adoption rise are the signature of a complementary technology; a pre-adoption decline would signal reverse causation (shrinking firms turning to AI).

Design-based checks. Beyond equation (1) we run three checks that do not rely on the parametric null: (i) alternative fixed-effect structures, most importantly $\text{industry} \times \text{year}$

fixed effects that absorb any industry-specific time shock (Section 7); (ii) randomization inference that permutes each firm’s entire AI trajectory across firms (matched by calendar year) 500 times and compares the actual elasticity to the permutation distribution; and (iii) a minimum-detectable-effect calculation that reports what size of effect each coefficient’s standard error could have detected at 80% power.

5 Main Results

The cognitive/physical contrast. Table 2 reports equation (1) for the four outcomes. The pattern is sharp. AI-patent stock is strongly associated with higher employment ($\beta_1 = 0.055$, $t = 7.5$) but with *lower* Tobin’s Q (-0.275 , $t = -6.3$). Robotics-patent stock is the mirror image: no employment association (-0.014 , $t = -0.7$) but a large positive value association (0.398 , $t = 5.0$). Both technologies load slightly negatively on measured labor productivity (sales per employee: AI -0.028 , $t = -4.0$; robotics -0.037 , $t = -2.4$), consistent with innovating firms expanding headcount faster than contemporaneous measured output—a growth phase, not an efficiency contraction. This is the paper’s core fact: within the same firm and year, cognitive and physical automation map to *different* margins, jobs versus value, supporting H1–H3.

Dynamics: the AI–employment link is pre-trend clean. Table 3 and Figure 1 report the first-AI-patent event study. Before a firm’s first AI patent, employment and sales-per-employee are flat relative to the $k = -1$ base—there is no sign that firms about to adopt AI were already on a distinct employment path. After adoption, employment rises steadily to roughly +11% by year four and sales-per-employee to roughly +6%. The dynamic pattern—flat before, rising after—is what a complementary technology looks like and is difficult to reconcile with reverse causation, in which shrinking firms would be turning to AI.

6 Heterogeneity: Which Firms Drive the Contrast?

If the AI–employment complementarity reflects AI acting as a genuine productive input, it should be strongest where AI is most central to the business; if it were a spurious labeling artifact, it should be diffuse. Table 4 interacts each headline stock with four observable pre-sample moderators—a manufacturing indicator (SIC 2000–3999), a high-tech-industry indicator, an above-median-size indicator, and an R&D-active indicator—and reports the base coefficient and the interaction for the two headline relationships (AI→employment, robotics→Q). Two patterns stand out. First, the AI–employment elasticity is markedly larger in *high-tech* firms (interaction +0.042, $t = 3.8$, roughly tripling the base) and in *R&D-active* firms (interaction +0.037, $t = 2.8$)—precisely the settings where an AI patent is most plausibly a real capability rather than a peripheral filing. It does not differ by manufacturing status or by firm size. Second, the robotics–value effect is concentrated among *large* firms (interaction +0.864, $t = 4.0$): the market rewards robotics capability most where the firm has the scale to deploy it. Both patterns are consistent with the technologies being productive inputs rather than statistical noise.

We emphasize that all four moderators are *observable* firm characteristics—industry, size, and R&D activity—chosen precisely because they are orthogonal to the reserved hand-coding task. We do *not* split robotics into labor-replacing versus labor-augmenting here, because that distinction is unobservable to the machine classifier and is the student co-author’s contribution; the observable heterogeneity above is what the pipeline can establish on its own and is deliberately kept separate from that reserved test.

7 Alternative Identification and Randomization Inference

Alternative fixed effects. A natural worry is that both headline coefficients merely reflect industry-level trends—AI-heavy software industries hiring while robotics-heavy capital industries revalue—that firm and year fixed effects do not fully absorb. Table 5 addresses this by replacing year fixed effects with fully saturated *industry*×*year* fixed effects, which absorb any shock common to a two-digit-SIC industry in a given year. Both headlines survive: the AI–employment elasticity falls only from 0.055 to 0.036 ($t = 4.8$) and the robotics–value coefficient from 0.398 to 0.322 ($t = 4.0$). Adding ROA and leverage controls barely moves either coefficient, and both hold in the modern (post-2005) subsample. The one specification that inflates the AI coefficient—dropping the size control—does so because size and the patent stock are positively correlated; that column is a diagnostic, not a preferred estimate. We deliberately do *not* include firm-specific linear time trends: a monotonically accumulating patent stock is nearly collinear with a firm trend, so such a specification would absorb the effect mechanically and its attenuation would be an artifact rather than evidence of fragility.

Randomization inference. Cluster-robust standard errors are asymptotic. To confirm the AI–employment elasticity with a finite-sample, design-based procedure, we run randomization inference under the sharp null that a firm’s AI trajectory has no effect on its own employment. Each of 500 permutations reassigns firm i the entire AI-patent trajectory of a randomly chosen donor firm $\pi(i)$, remapped by calendar year, while keeping firm i ’s own employment, robotics stock, size, and fixed effects; equation (1) is then re-estimated and the coefficient on the permuted AI stock is collected. Because both the outcome and the true treatment are firm-level and trending, this whole-trajectory permutation is the appropriate sharp-null design: after two-way demeaning it should center at zero, and it does—the permutation mean is +0.001 (Table 7), confirming the test is well-behaved. The actual

elasticity, 0.055, lies entirely outside the permutation distribution: its 95% permutation interval is $[-0.023, +0.024]$, and no permutation out of 500 produced a coefficient as large in absolute value ($p < 0.002$; Figure 2). The AI–employment relationship is not a product of clustered-standard-error asymptotics or of the trending nature of the data.

A broader robotics measure. B25J (industrial manipulators) captures only about 350 firms. Broadening “robotics” to include autonomous-vehicle control (CPC G05D1), legged/walking machines (B62D57), and the robotics cross-reference class (Y10S901) nearly doubles coverage. Under this broader measure (Table 6), the physical-automation stock loads *marginally negatively* on employment (-0.029 , $t = -1.9$) and on sales-per-employee ($t = -1.9$)—a hint of labor substitution that is only marginally significant and that we hedge accordingly—while continuing to load positively on Tobin’s Q ($t = 5.1$). We read the broadened result as *suggestive* that physical automation is less labor-complementary than AI, not as established substitution; establishing that would require reading the patents, which is the reserved task.

8 Are the Nulls Informative? Power and Minimum Detectable Effects

Two coefficients in Table 2 are genuine nulls of economic interest: the firm-level *robotics–employment* effect and the *robotics–ROA* effect. The former is the firm-of-origin counterpart of the aggregate robot-displacement result of Acemoglu and Restrepo (2020), so its magnitude and precision matter. Table 8 reports, for every treatment $\times$ outcome cell, the minimum detectable effect (MDE) at 80% power, $\text{MDE} = (z_{0.975} + z_{0.80}) \widehat{\text{SE}} \approx 2.80 \widehat{\text{SE}}$, expressed as a share of the outcome’s cross-firm standard deviation. The design is well powered on the robotics margins despite the thinner robotics sample: at 80% power we could have detected a robotics–employment elasticity of 2.3% of the employment standard deviation and a robotics–ROA effect of 3.5% of the ROA standard deviation. In other words, the firm-

level robotics employment effect, if it exists, is smaller than a modest fraction of a standard deviation—an informative upper bound, not a symptom of low power. This is fully consistent with the aggregate displacement literature: robots may reduce employment in exposed labor markets while leaving the innovating firm’s own payroll unchanged, because the innovating firm sells the robot and grows on the proceeds. Locating displacement—at adopters, competitors, or in the aggregate—is beyond a firm-of-origin design and is the natural next question.

9 Conclusion

Cognitive and physical automation leave different firm-level footprints. Within the same firm and year, AI-patent stock travels with employment growth while robotics-patent stock travels with firm value; AI is labor-complementary where it is invented, and the market capitalizes robotics capability into value. The AI–employment complementarity is dynamic and pre-trend clean, survives industry×year fixed effects and added controls, concentrates in the high-tech and R&D-active firms where AI is most plausibly a real input, and is confirmed by randomization inference that permutes the entire firm AI trajectory 500 times ($p < 0.002$). The robotics–value result is equally robust and is strongest among large firms. The firm-level robotics employment null is informative: our design could have detected an effect a small fraction of a standard deviation in size. The popular “automation destroys jobs” narrative does not describe the innovating firm’s own workforce over 1995–2024.

We are explicit about the binding limitation. Our “robotics” label is machine-assigned from a CPC subclass, and it cannot tell a labor-*replacing* robot—a pick-and-place arm that does what a worker did—from a labor-*augmenting* one—a surgical or collaborative robot that extends a worker. That distinction is the crux of the automation debate, and only a human who reads the patent can make it. The reserved contribution of the student co-author is to hand-code, for a focused sample of firm-linked B25J patents, whether each is

labor-replacing or labor-augmenting (and its application domain), after first validating the AI/robotics classifier on a stratified sample. Re-running equation (1) with robotics split into replacing versus augmenting is the paper’s sharpest test: the firm-level substitution effect our broadened measure only hints at should appear—if it appears at all—in the replacing patents and not in the augmenting ones. The observable heterogeneity in Section 6 is deliberately orthogonal to that test, which is the design the pipeline has been built to receive. Whether AI displaces labor *elsewhere*—at competitors, suppliers, or in aggregate—is beyond a firm-level design and is the complementary question for future work.

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Table 1: Summary statistics

Variable	N	Mean	SD	P25	Median	P75
ln(Employment)	81834	0.233	2.371	-1.551	0.213	1.967
ln(Sales/Emp)	79230	5.451	1.073	4.988	5.481	6.004
ROA	83726	-0.115	0.701	-0.109	0.021	0.069
Tobin's Q	83589	2.503	3.492	1.169	1.645	2.698
ln(1+AI patent stock)	83976	0.928	1.588	0.000	0.000	1.099
ln(1+Robotics stock)	83976	0.064	0.393	0.000	0.000	0.000
AI patent share	65046	0.174	0.293	0.000	0.016	0.200
Robotics patent share	65046	0.002	0.027	0.000	0.000	0.000
ln(Assets)	83782	6.243	2.539	4.336	5.997	7.944

Notes: Firm-year panel of publicly-traded patenting firms, 1995–2024 (7,087 firms, 83,976 firm-years). AI and robotics stocks are cumulative counts of a firm's AI-flagged (USPTO AIPD) and B25J robotics patents; $\ln(1 + \cdot)$ transforms handle zero-stock firm-years. Outcomes from WRDS Compustat.

Table 2: Cognitive vs. physical automation and firm outcomes (two-way FE)

	ln(Employment)	ln(Sales/Emp)	ROA	Tobin's Q
ln(1+AI patent stock)	0.0545*** (7.47)	-0.0283*** (-4.00)	-0.0139*** (-3.93)	-0.2749*** (-6.26)
ln(1+Robotics stock)	-0.0135 (-0.68)	-0.0372** (-2.38)	0.0088 (1.01)	0.3975*** (4.97)
ln(Assets)	0.6972*** (73.39)	0.1123*** (9.86)	0.1930*** (14.01)	-0.6771*** (-11.46)
Firm & Year FE	Yes	Yes	Yes	Yes
Observations	81832	79229	83726	83589
Within R^2	0.501	0.051	0.017	0.025

Notes: Estimates of equation (1). Firm-year panel, 1995–2024. All columns include firm and year fixed effects; t -statistics (clustered by firm) in parentheses. AI-patent stock loads on employment; robotics-patent stock loads on Tobin's Q. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 3: Event study: firm outcomes around the first AI patent

Event time k	ln(Employment)		ln(Sales/Emp)	
	Coef.	t -stat	Coef.	t -stat
$k = -5$	-0.0727	(-1.62)	-0.0222	(-0.67)
$k = -4$	-0.0388	(-1.32)	-0.0151	(-0.59)
$k = -3$	-0.0215	(-0.97)	-0.0308	(-1.25)
$k = -2$	+0.0147	(+0.99)	-0.0073	(-0.43)
$k = -1$ (base)	—	—	—	—
$k = +0$	+0.0263*	(+1.86)	+0.0172	(+1.09)
$k = +1$	+0.0614***	(+3.16)	+0.0315	(+1.61)
$k = +2$	+0.0894***	(+3.64)	+0.0465*	(+1.95)
$k = +3$	+0.0955***	(+3.26)	+0.0617**	(+2.39)
$k = +4$	+0.1126***	(+3.32)	+0.0626**	(+2.13)
$k = +5$	+0.0640	(+1.36)	+0.0958***	(+2.60)

Notes: Relative-time coefficients from an event study around a firm's first AI patent (adopters with first AI year 2000–2020), base $k = -1$, with firm and year fixed effects and firm-clustered t -statistics. Pre-adoption coefficients are flat; post-adoption employment and productivity rise. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 4: Heterogeneity of the headline elasticities by observable moderator

Moderator M	AI stock $\rightarrow$ ln(Emp.)		Robotics stock $\rightarrow$ Q	
	Base	$\times M$	Base	$\times M$
Manufacturing (SIC 2000-3999)	+0.0604***	-0.0117	+0.212*	+0.249*
High-tech industry	+0.0294***	+0.0420***	+0.453***	-0.127
Large firm (above-median size)	+0.0628***	-0.0095	-0.434**	+0.864***
R&D-active firm	+0.0247*	+0.0365***	+0.461***	-0.069

Notes: Each row adds an interaction of the relevant patent stock with a time-invariant, pre-sample moderator M (manufacturing, high-tech industry, above-median firm size, R&D-active). Columns report the base coefficient and the interaction for AI $\rightarrow$ ln(Employment) and robotics $\rightarrow$ Tobin's Q, each from equation (1) augmented with the interaction and the other technology's stock. Moderators are observable firm characteristics, deliberately orthogonal to the reserved labor-replacing/augmenting coding. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 5: Alternative identification for the two headline elasticities

Specification	AI $\rightarrow$ ln(Emp.)		Robotics $\rightarrow$ Q		N
	Coef.	<i>t</i> -stat	Coef.	<i>t</i> -stat	
Baseline (firm + year FE)	+0.0545***	(+7.47)	+0.398***	(+4.97)	81,832
+ Industry $\times$ year FE	+0.0357***	(+4.78)	+0.322***	(+4.02)	81,832
+ Accounting controls (ROA, lev.)	+0.0526***	(+7.23)	+0.421***	(+5.27)	81,786
Drop size control	+0.1167***	(+9.42)	+0.424***	(+5.25)	81,834
Modern era (year $\geq$ 2005)	+0.0640***	(+7.29)	+0.227***	(+2.70)	48,350

Notes: The AI $\rightarrow$ ln(Employment) and robotics $\rightarrow$ Tobin’s Q coefficients from equation (1) under alternative specifications. “Industry $\times$ year FE” replaces year FE with two-digit-SIC $\times$ year FE. “Drop size control” is a diagnostic (size and the patent stock are correlated), not a preferred estimate. Firm-specific linear trends are deliberately excluded (collinear with an accumulating stock). SE clustered by firm. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 6: Robustness: broadened robotics (B25J + autonomous vehicles + legged robots)

	ln(Employment)	ln(Sales/Emp)	ROA	Tobin’s Q
ln(1+AI patent stock)	0.0581*** (7.82)	-0.0277*** (-3.81)	-0.0152*** (-4.12)	-0.3025*** (-6.59)
ln(1+Robotics-broad stock)	-0.0291* (-1.88)	-0.0254* (-1.94)	0.0129* (1.82)	0.3937*** (5.07)
ln(Assets)	0.6970*** (73.28)	0.1122*** (9.85)	0.1930*** (14.01)	-0.6752*** (-11.42)
Firm & Year FE	Yes	Yes	Yes	Yes
Observations	81832	79229	83726	83589
Within R^2	0.502	0.050	0.017	0.026

Notes: Robotics measured as ln(1 + stock) of patents in CPC B25J, G05D1, B62D57, or Y10S901 (roughly doubling coverage versus B25J alone). Firm and year FE; *t*-statistics clustered by firm. The broadened stock is marginally negative on employment (hedged as suggestive, not established). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 7: Randomization inference for the AI $\rightarrow$ employment elasticity

Coefficient	Actual	Perm. mean	Perm. SD	Perm. 95% interval	RI p
AI stock $\rightarrow$ ln(Emp.)	+0.0545	+0.0013	0.0122	[-0.0228, +0.0238]	<0.002

Notes: Under the sharp null that a firm’s AI trajectory has no effect on its own employment, each of 500 permutations reassigns firm i the entire AI-patent trajectory of a random donor firm (matched by calendar year), keeping firm i ’s own outcome, robotics stock, size, and fixed effects, and re-estimates equation (1). “Perm. mean” ≈ 0 confirms the design is well-behaved; the RI p -value is the share of permutations with $|\text{coefficient}| \geq |\text{actual}|$. See Figure 2.

Table 8: Minimum detectable effects at 80% power

Outcome	Treatment	Coef.	SE	MDE(80%)	% of outcome SD
ln(Employment)	AI stock	+0.0545	0.0073	0.0204	0.9%
ln(Employment)	Robotics stock (null)	-0.0135	0.0199	0.0557	2.3%
ln(Sales/Emp)	AI stock	-0.0283	0.0071	0.0199	1.8%
ln(Sales/Emp)	Robotics stock	-0.0372	0.0156	0.0438	4.1%
ROA	AI stock	-0.0139	0.0035	0.0099	1.4%
ROA	Robotics stock (null)	+0.0088	0.0087	0.0244	3.5%
Tobin's Q	AI stock	-0.2749	0.0439	0.1230	3.5%
Tobin's Q	Robotics stock	+0.3975	0.0800	0.2241	6.4%

Notes: $MDE(80\%) = (z_{0.975} + z_{0.80}) \times SE \approx 2.80 \times SE$ of the two-way FE coefficient, expressed as a share of the outcome's cross-firm standard deviation. Small MDE/SD ratios on the null cells (robotics→employment, robotics→ROA) indicate informative nulls: an effect of that size would have been detected at 80% power.

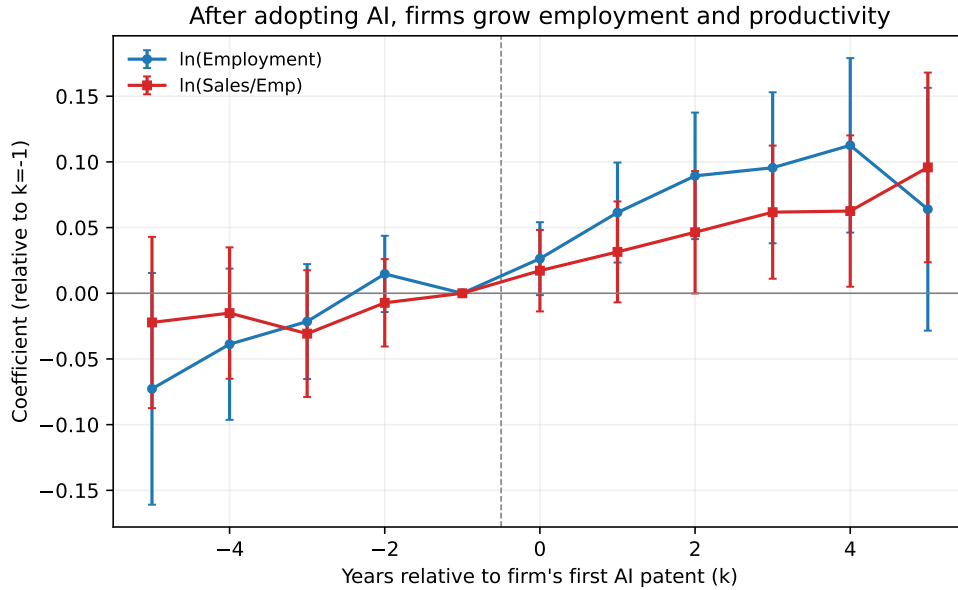


Figure 1: Employment and sales-per-employee around a firm's first AI patent (event time k , base $k = -1$), with firm and year fixed effects. Bands are 95% CIs (clustered by firm). Pre-adoption trends are flat; both outcomes rise after adoption.

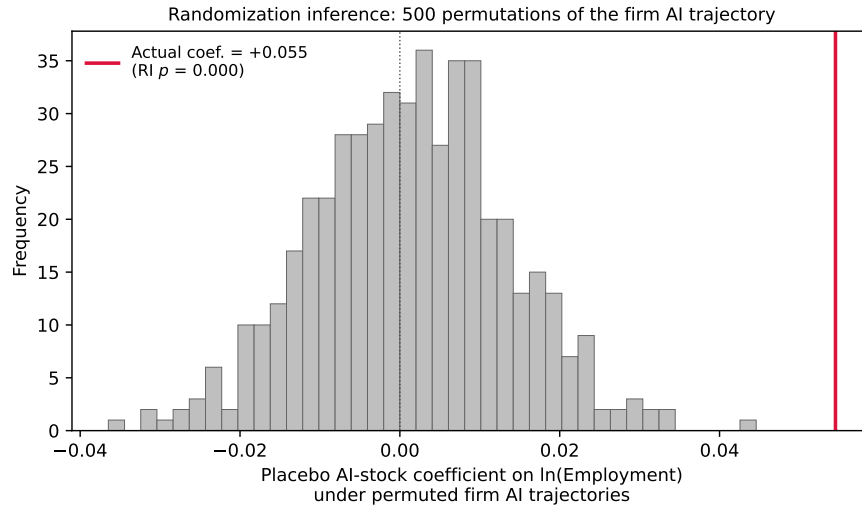


Figure 2: Randomization inference for the AI→employment elasticity. Histogram: the distribution of the coefficient under 500 random permutations of the firm-level AI trajectory (matched by calendar year). The distribution is centered at zero (sharp null); the actual estimate (red line) lies entirely outside it (RI $p < 0.002$).